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5.5.4 Kinetic Energy in Rotation

Let us consider a rigid body rotating with angular velocity ω about an axis as shown in Figure 5.30. Every particle of the body will have the same angular velocity ω and different tangential velocities v based on its positions from the axis of rotation.

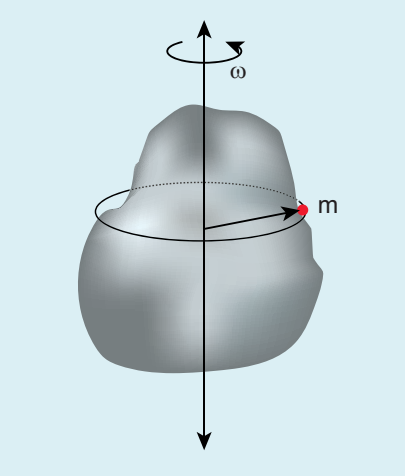


Figure 5.30 Kinetic energy in rotation

Let us choose a particle of mass m_i situated at distance r_i from the axis of rotation. It has a tangential velocity v_i given by the relation, $v_i = r_i \omega$. The kinetic energy KE_i of the particle is,

$$KE_i = \frac{1}{2} m_i v_i^2$$

Writing the expression with the angular velocity,

$$KE_i = \frac{1}{2} m_i (r_i \omega)^2 = \frac{1}{2} (m_i r_i^2) \omega^2$$

For the kinetic energy of the whole body, which is made up of large number of such particles, the equation is written with summation as,

$$KE = \frac{1}{2} \left(\sum m_i r_i^2 \right) \omega^2$$

where, the term $\sum m_i r_i^2$ is the moment of inertia I of the whole body. $I = \sum m_i r_i^2$

Hence, the expression for KE of the rigid body in rotational motion is,

$$KE = \frac{1}{2} I \omega^2 \qquad (5.52)$$

This is analogous to the expression for kinetic energy in translational motion.

$$KE = \frac{1}{2} M v^2$$

Relation between rotational kinetic energy and angular momentum

Let a rigid body of moment of inertia I rotate with angular velocity ω.

The angular momentum of a rigid body is, $L = I \omega$

The rotational kinetic energy of the rigid body is, $KE = \frac{1}{2} I \omega^2$

By multiplying the numerator and denominator of the above equation with I, we get a relation between L and KE as,

$$KE = \frac{1}{2} \frac{I^2 \omega^2}{I} = \frac{1}{2} \frac{(I\omega)^2}{I}$$

$$KE = \frac{L^2}{2I} \qquad (5.53)$$

EXAMPLE 5.20

Find the rotational kinetic energy of a ring of mass 9 kg and radius 3 m rotating with 240 rpm about an axis passing through its center and perpendicular to its plane. (rpm is a unit of speed of rotation which means revolutions per minute)